

Does AI Adoption Improve Institutional Performance? Evidence from Colombian Higher Education

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Abstract

Higher education institutions across Latin America are adopting artificial intelligence tools for administrative and governance functions at an accelerating pace, yet the causal effects of this adoption on institutional performance remain unexamined. While the existing literature documents AI applications in university management and proposes governance frameworks, the intersection between AI adoption, institutional efficiency, and student outcomes has received no rigorous empirical treatment in the Latin American context. This paper lays the groundwork for a causal evaluation of AI adoption in Colombian higher education institutions. We review the relevant literature, characterize the Colombian institutional context and its suitability for identification, describe a proposed staggered difference-in-differences strategy using SNIES panel data combined with a primary adoption survey, and present a calibration exercise that derives expected effect ranges from comparable international evidence. The calibration suggests that AI adoption in administrative functions may reduce first-year dropout rates by 1.5 to 5 percentage points and administrative costs by 8 to 18 percent, with effects concentrated among better-resourced institutions. These projections motivate the full empirical agenda.

Keywords: artificial intelligence, higher education, institutional efficiency, Colombia, causal identification, staggered difference-in-differences

JEL codes: I23, O33, J24, C23

1. Introduction

The governance of higher education institutions is information-intensive by design. Admissions decisions, resource allocation, student retention interventions, and academic planning all require processing large volumes of data to generate timely and accurate outputs. Artificial intelligence — computational systems that use data to perform tasks previously requiring human judgment — promises to transform each of these functions. Over the past decade, universities in the United States, Europe, and increasingly Latin America have deployed machine learning tools for dropout prediction, natural language processing systems for student advising, and data analytics platforms for strategic planning and administrative management (George et al., 2023; Salas-Pilco et al., 2022).

The efficiency case for AI in university administration is straightforward. Administrative processes in higher education are labor-intensive, slow, and prone to inconsistency. A system that flags at-risk students before they disengage, processes financial aid applications in hours rather than weeks, or identifies budget inefficiencies across academic units creates measurable value. In Colombia, first-year dropout rates in higher education exceeded 13 percent in 2022, and administrative expenditure per enrolled student varies by a factor of four across the public university system (SNIES, 2023). At this scale, even modest efficiency improvements translate into substantial social returns.

Despite the growth of both the technology and the policy interest, the evidence base for AI adoption effects on institutional outcomes remains thin. A systematic review of the Latin American literature finds that most published work takes the form of descriptive surveys, taxonomy proposals, or qualitative case studies (Salas-Pilco et al., 2022). Causal estimates of the effect of AI adoption on institutional performance are absent from the regional literature. The most empirically ambitious contribution in the context — Nieto et al. (2019) — uses machine learning to predict graduation rates at South American engineering programs, but this is a predictive exercise. It documents that AI can forecast outcomes; it does not establish whether AI adoption causes improvements in those outcomes.

The identification problem is fundamental. Institutions that adopt AI tools for administration differ systematically from those that do not: they tend to be better resourced, more efficiently managed, and better positioned to capture efficiency gains. Any comparison of outcomes between adopters and non-adopters that does not account for this selection will overstate the causal effect of adoption. Resolving this problem requires a research design that generates variation in AI adoption that is plausibly independent of the pre-existing institutional efficiency trajectory.

This paper addresses that design challenge directly. We argue that the Colombian higher education system offers the conditions necessary for credible causal identification — a heterogeneous panel of institutions observed over time in SNIES, variation in AI adoption timing driven in part by accreditation cycle pressures and national digital transformation policy, and a primary survey instrument capable of recovering adoption timing at the institution level. We lay out the full empirical strategy, derive expected effect ranges from the international evidence, and propose the data collection agenda needed to execute it.

Three contributions distinguish this paper from the existing literature. First, relative to the descriptive LAC literature (Salas-Pilco et al., 2022; George et al., 2023), we provide the first identification framework for estimating causal effects of AI adoption on institutional outcomes in the region. Second, relative to Nieto et al. (2019), whose predictive approach cannot separate AI tool effects from institutional selection, we propose a staggered DiD design that exploits variation in adoption timing to isolate the causal effect. Third, we present a structured calibration exercise that translates international effect estimates into plausible LAC-specific ranges, providing the empirical priors necessary to power the survey and evaluation design.

The rest of the paper is organized as follows. Section 2 describes the Colombian higher education system and its suitability for identification. Section 3 reviews the relevant literature and maps the central tensions in the field. Section 4 develops the proposed empirical strategy. Section 5 presents the calibration exercise and its implications for the full analysis. Section 6 concludes.

2. The Colombian Higher Education System: Institutional Context

Colombia's higher education system is one of the most institutionally diverse in Latin America. As of 2023, SNIES registers approximately 296 active higher education institutions (HEIs), encompassing public and private universities, university institutions, technological institutions, and technical professional schools. Total enrollment exceeds 2.4 million students, of whom approximately 52 percent attend private institutions. The system spans a wide range of institutional capacity: flagship public universities such as the Universidad Nacional de Colombia operate with annual budgets exceeding COP 1.5 trillion, while small regional technical institutions manage with budgets two orders of magnitude smaller (MEN, 2023).

This institutional diversity is not merely descriptive background. It is the source of the identifying variation that the empirical strategy requires. Institutions with different resource levels, accreditation statuses, and administrative capacities adopted AI tools for administrative and governance functions at different times and for different reasons. The staggered timing of this adoption — driven in part by factors that are plausibly independent of pre-existing efficiency trajectories — is the foundation of the causal identification strategy.

2.1 SNIES and the Data Infrastructure

The SNIES, administered by the Ministerio de Educación Nacional (MEN), provides annual institution-level data on enrollment, graduation and dropout rates, financial indicators (budget, administrative expenditure, sources of funding), and staff composition (administrative and academic, by contract type). Coverage is comprehensive: all formally registered HEIs are required to report to SNIES as a condition of operation. Data are available from 2009 onward, generating panels of 14 or more years for most institutions currently in operation.

For the purposes of causal identification, SNIES offers three outcome variables of direct interest: the first-year dropout rate, which captures the most immediate and costly form of student attrition; administrative expenditure per enrolled student, which measures operational efficiency in resource use; and the administrative staff-to-enrollment ratio, which captures the labor market channel through which AI adoption may reduce employment in support and clerical functions. Each of these outcomes is measured annually, consistently defined across institutions, and available for the full panel period.

SNIES does not currently include information on AI adoption. This is the binding data constraint that the primary survey instrument is designed to address. The survey collects, for each responding institution: the year of first deployment of any AI-assisted administrative tool; the specific functions in which AI tools are deployed (admissions, student retention, financial management, academic analytics, strategic planning); the vendor or platform used; and a self-assessed measure of deployment depth (pilot, selective deployment, institution-wide integration). The combination of SNIES outcomes and survey-derived adoption timing constitutes the core dataset for the causal analysis.

2.2 Policy Context and Sources of Adoption Timing Variation

Variation in AI adoption timing across Colombian HEIs is generated by at least three sources, each of which creates plausible exogenous variation that the identification strategy can exploit.

First, the accreditation cycle administered by the Consejo Nacional de Acreditación (CNA) creates periodic pressure for institutions to upgrade administrative systems. Institutions entering their accreditation review cycle face heightened scrutiny of data management practices, information systems, and evidence-based decision-making capacity. The timing of accreditation cycles varies across institutions for historical reasons largely unrelated to pre-existing efficiency levels, generating variation in the pressure to adopt administrative AI tools that is partially independent of the institution's prior trajectory.

Second, CONPES 3975 of 2019, Colombia's national digital transformation policy, established AI adoption as a priority for public sector institutions, including higher education. The policy created financial incentives — through Minciencias and MEN programs — for specific institutions to invest in digital administrative infrastructure in 2020 and 2021. Institutions that received these grants adopted AI tools earlier than comparable institutions that did not, providing a second source of quasi-exogenous timing variation.

Third, geographic proximity to technology hubs — Bogotá, Medellín, and Cali — is associated with earlier exposure to software vendors and lower transaction costs for AI platform adoption. Distance from these hubs generates spatial variation in adoption timing that is plausibly independent of institution-level efficiency trends, conditional on urbanization and size controls.

No single source of variation is sufficient for identification on its own. The empirical strategy uses the combination of these three sources — accreditation cycle timing, CONPES grant receipt, and proximity to technology infrastructure — both as the basis for pre-trend validation and, in robustness specifications, as instruments for adoption timing.

3. Literature Review: Opportunities, Evidence, and Tensions

The literature on AI in higher education governance organizes into three bodies of work: a global literature on efficiency opportunities and administrative applications; a critical literature on ethical risks and equity implications; and a regional LAC literature that documents adoption patterns without producing causal estimates. We review each in turn, then map the central tensions that a causal evaluation of AI adoption must address.

3.1 Efficiency Opportunities

The efficiency case for AI in university administration rests on three mechanisms. The first is early identification of student dropout risk. Machine learning models trained on enrollment, performance, and socioeconomic data identify at-risk students with substantially greater accuracy than conventional administrative monitoring. Nieto et al. (2019) demonstrate that random forest classifiers outperform logistic regression and decision tree alternatives in predicting graduation rates at South American engineering programs. George et al. (2023), reviewing smart university

deployments globally, report reductions in first-year dropout rates of 2 to 8 percentage points in controlled evaluations, though these estimates conflate AI tool effects with selection into adoption.

The second mechanism is process automation in back-office functions. Natural language processing systems handle routine student inquiries and document classification. Robotic process automation reduces the labor cost of repetitive data entry, compliance reporting, and records management. Khairullah et al. (2025) document that automation reallocates administrative labor from routine tasks toward student support and strategic functions, with self-reported cost reductions of 15 to 30 percent in the functions where implementation occurs. Vorontsova et al. (2025), in a bibliometric analysis of 234 articles on AI economic effects in higher education, identify administrative cost reduction, staff time savings, and improvements in service accessibility as the three most consistently documented outcomes.

The third mechanism is decision support in institutional governance. Data analytics platforms aggregate enrollment, financial, and research data to support leadership decisions on resource allocation, program portfolio management, and strategic planning. Spósito (2025) develops a taxonomy of AI applications in HE leadership across ten domains, with governance and compliance and decision-making and policy formulation among the fastest-growing application areas between 2017 and 2024.

3.2 Risks and the Critical Literature

The efficiency literature understates three categories of risk that qualify the efficiency gains. The first is algorithmic bias. AI systems trained on historical institutional data reproduce the inequities embedded in that data. Dropout prediction models that overestimate risk for students from lower socioeconomic backgrounds trigger misallocated interventions — or denied resources — that reinforce existing disadvantage. Wang (2020) documents this mechanism in the educational leadership context, arguing that AI-driven data-informed decision-making amplifies rather than corrects historical biases because the training data reflects patterns of past discrimination. Kasy and Abebe (2021) demonstrate formally that standard algorithmic fairness metrics legitimate inequality by conditioning on measured characteristics while ignoring unmeasured background factors.

The second risk is data governance failure. AI systems for university administration require access to extensive personal data — academic records, financial information, behavioral data from learning management systems — and the governance of this data in the LAC context is inadequately regulated. Filgueiras (2023) argues that AI-driven governance practices institutionalize new power asymmetries within educational institutions, concentrating data authority in administrative and technical units and reducing the democratic character of institutional governance. Most LAC countries lack personal data legislation for educational contexts equivalent to the European GDPR or the US FERPA, meaning that AI adoption proceeds without the legal infrastructure that constrains it in higher-capacity jurisdictions.

The third risk is accountability erosion. Decision-support systems that provide algorithmic recommendations shift the burden of justification in high-stakes decisions. Wang (2020) frames this as a conflict between computational efficiency and the moral values — fairness, transparency, individual dignity — that educational institutions are formally committed to upholding. Kern et al. (2025) make a related methodological argument: reliable algorithmic decision-making requires

explicit causal reasoning about the relationship between inputs and outcomes, not merely predictive accuracy. Systems that perform well on average may fail precisely in the cases where human contextual judgment is most valuable.

3.3 The Latin American Evidence

The LAC-specific literature documents adoption patterns, identifies barriers, and catalogs governance concerns, but does not produce causal evidence on outcomes. Salas-Pilco et al. (2022), in the most comprehensive systematic review of AI in LAC higher education, find that AI applications are concentrated in a small number of well-resourced institutions, primarily in Brazil, Mexico, Colombia, and Chile, and that administrative applications are less developed than pedagogical ones. García Peñaloza et al. (2025) document five dominant AI management strategies in the region — predictive student analytics, automated administrative workflows, data-driven resource allocation, AI-assisted accreditation management, and digital communication platforms — with adoption rates varying substantially by institution type and size.

Rivadeneira et al. (2024), in qualitative interviews across Ecuador, Chile, and Costa Rica, find that AI's role in formal institutional decision-making remains limited and that decision-makers identify three unresolved challenges: distinguishing human from AI-generated analysis, maintaining critical thinking in leadership processes, and managing the student inequalities that AI systems expose. Velázquez Avila et al. (2025) document that digitalization in LAC public universities advances faster in administrative functions than in pedagogical ones, frequently without ethical frameworks or faculty participation — a pattern consistent with adoption driven by efficiency pressure rather than deliberate institutional governance.

3.4 The Central Tension: Efficiency-Equity in Unequal Systems

The review reveals a central tension that is sharper in LAC than in the global literature. Well-resourced institutions — which are most capable of implementing AI governance tools effectively — serve predominantly higher-income students. Under-resourced public universities — where the efficiency gains from AI would generate the greatest social return, and where dropout rates are highest — lack the data infrastructure to capture those gains. This distributional pattern implies that unrestricted adoption incentives will widen the performance gap between the top and bottom of the Colombian higher education system. The heterogeneity analysis in the proposed empirical design is directly motivated by this concern.

4. Proposed Empirical Strategy

We propose to estimate the causal effect of AI adoption in administrative and governance functions on institutional outcomes in Colombian HEIs using a staggered difference-in-differences design. The design exploits variation in AI adoption timing across institutions in the SNIES panel, using the primary survey instrument described in Section 2 to recover adoption dates. This section develops the estimand, identification assumption, implementation, and the principal threat to identification.

4.1 Estimand

The parameter of interest is the average treatment effect on the treated (ATT): the expected change in institutional outcome Y for Colombian HEIs that adopted AI in administrative functions, relative to the counterfactual outcome those institutions would have experienced absent adoption. This is a causal parameter, not a descriptive comparison. It answers the policy-relevant question — does AI adoption improve outcomes for adopting institutions? — while remaining agnostic about what would happen to non-adopters if induced to adopt.

We estimate the ATT for three outcomes: the first-year dropout rate; administrative expenditure per enrolled student; and the administrative staff-to-enrollment ratio. Each outcome captures a distinct mechanism through which AI adoption may affect institutional performance. The dropout rate captures the student retention channel. Administrative expenditure captures the process efficiency channel. The staff-to-enrollment ratio captures the labor market channel — whether AI substitutes for administrative labor, complements it, or has no detectable employment effect at the institution level.

4.2 Identification Strategy

To identify the causal effect of AI adoption on institutional outcomes, we use a staggered difference-in-differences strategy that exploits the spatial and temporal variation in adoption timing across Colombian HEIs. This strategy recovers the ATT from the deviation of permanent differences between treatment and control groups following adoption, under the assumption that these differences would have remained stable absent treatment.

Formally, let Y_{it} denote the outcome for institution i in year t , and let $D_{it} = 1$ if institution i has adopted AI in administrative functions by year t . The staggered DiD estimator follows Callaway and Sant'Anna (2021), which computes group-time ATTs for each combination of adoption cohort g and post-adoption period t , then aggregates them into an overall ATT estimate. This approach avoids the negative weighting problem in the conventional two-way fixed effects estimator, which can produce sign-reversed estimates when treatment effects are heterogeneous across cohorts — a concern directly relevant to this setting, where efficiency gains are expected to vary substantially by institution type.

The core identification assumption is conditional parallel trends: conditional on observable institutional characteristics (size, public/private status, CNA accreditation level, prior administrative expenditure trajectory), institutions that adopted AI in different years would have followed parallel outcome trends absent adoption. This assumption is testable using the pre-adoption period in SNIES, which provides between six and twelve years of pre-treatment data for most institutions currently operating. We assess the plausibility of parallel trends by regressing outcomes on cohort-specific pre-trends and testing for statistically significant divergence before adoption.

4.3 Main Threat and Robustness

The principal threat to identification is selection on pre-existing efficiency trajectories. Institutions that were already improving in administrative efficiency before adoption may have

been more likely to adopt early, generating spurious positive estimates even after controlling for observable characteristics. Three robustness checks address this threat directly.

First, we restrict the comparison group to institutions in the same CNA accreditation cohort as each adopting institution. Institutions that entered CNA review in the same cycle faced similar external pressure to upgrade administrative systems; comparing adopters and non-adopters within the same accreditation cohort reduces the selection bias from efficiency-driven adoption.

Second, we exclude institutions that received Minciencias digitalization grants in 2020–2021 from the main analysis and examine them separately. These institutions adopted AI in part because of external resource injections that may have also improved outcomes through channels unrelated to the AI tool itself — confounding the effect of adoption with the effect of funding.

Third, we use geographic proximity to technology hubs as an instrument for adoption timing in an IV specification, under the assumption that distance to Bogotá, Medellín, and Cali affects AI adoption timing through vendor access and transaction costs but does not directly affect institutional outcomes conditional on controls. This specification provides a bound on the selection bias in the DiD estimates.

4.4 Data Requirements and Survey Design

The proposed analysis requires two data sources. The SNIES panel provides outcomes and institutional controls for the full population of Colombian HEIs from 2009 to the present. The primary survey provides AI adoption timing and scope for the estimation sample. The survey is administered to administrative and IT directors at all HEIs registered in SNIES with ten or more years of continuous enrollment data — approximately 180 institutions. The survey instrument collects the year of first AI deployment by administrative function, the specific tool or vendor, and a self-assessed depth measure. Non-response follow-up and administrative record triangulation are specified in the survey protocol to reduce attrition bias in the adoption timing variable.

5. Calibration Exercise and Expected Results

***Note:** This section presents a calibration exercise, not empirical findings. The ranges reported below are derived from comparable international studies adjusted for LAC institutional conditions. They motivate the survey and evaluation design and establish the power requirements for the primary data collection. They should not be cited as estimates of the effect of AI adoption in Colombia.*

5.1 Calibration Approach

In the absence of primary survey data, we derive expected effect ranges from the closest international comparators. The calibration proceeds in three steps. First, we identify the studies in the literature that produce quantitative estimates most comparable to the parameters we propose to estimate — focusing on dropout rates, administrative costs, and staff composition in institutions comparable in resource level and institutional capacity to Colombian HEIs. Second, we apply a

downward adjustment factor to account for the lower data infrastructure quality and administrative baseline in the LAC context, following the adjustment logic in Salas-Pilco et al. (2022) and Chamba-Eras et al. (2026). Third, we estimate heterogeneity ranges based on the private/public and size dimensions that the literature identifies as the most important sources of differential effects.

The calibration does not assume that Colombian HEIs will achieve the same efficiency gains as North American or European counterparts. The adjustment factors are conservative: they assume that LAC institutions capture 50 to 70 percent of the efficiency gains documented in high-resource settings, reflecting the documented gaps in data quality, administrative capacity, and AI governance infrastructure.

5.2 Calibrated Effect Ranges

Table 1 summarizes the calibrated effect ranges for each outcome, alongside the international comparator and the basis for the LAC adjustment.

Table 1. Calibrated Expected Effects of AI Adoption on Institutional Outcomes in Colombian HEIs

Outcome	Comparable study	Estimate in source	Calibrated range (LAC)	Notes
First-year dropout rate (p.p.)	George et al. (2023)	−2 to −8 p.p.	−1.5 to −5 p.p.	Lower bound reflects data quality constraints in LAC
Admin cost per enrolled student (%)	Khairullah et al. (2025)	−15 to −30%	−8 to −18%	Adjusted for lower automation baseline in LAC HEIs
Admin staff-to-enrollment ratio (p.p.)	No direct LAC comparator	—	−0.5 to −2 p.p.	Range derived from process automation literature
Effect size, private vs. public institutions	Wang (2020); Salas-Pilco et al. (2022)	Private > public	Private 2–3× public	Driven by data infrastructure differential

Note: Calibrated ranges are derived from international comparators adjusted for LAC institutional conditions. They are not estimates from Colombian data. "p.p." denotes percentage points. Sources: George et al. (2023); Khairullah et al. (2025); Nieto et al. (2019); Salas-Pilco et al. (2022).

For the first-year dropout rate, the international literature documents reductions of 2 to 8 percentage points in controlled evaluations (George et al., 2023). Applied to the Colombian

baseline dropout rate of 13.2 percent (SNIES, 2023), a LAC-adjusted effect of 1.5 to 5 percentage points would imply between 11,000 and 37,000 fewer first-year dropouts annually across the Colombian system — a substantial social return even at the lower bound.

For administrative expenditure per enrolled student, studies of AI-assisted process automation in comparable institutional contexts report cost reductions of 15 to 30 percent in the functions where implementation occurs (Khairullah et al., 2025). Adjusted for the lower automation baseline in LAC HEIs, the calibrated range is 8 to 18 percent. Applied to median administrative expenditure per student in Colombian public universities of approximately COP 1.8 million (SNIES, 2023), this implies savings of COP 144,000 to COP 324,000 per enrolled student annually.

For the administrative staff-to-enrollment ratio, no direct LAC comparator exists. The calibration draws on the broader process automation literature, which documents employment reductions of 0.5 to 3 percentage points in the ratio of support staff to output in service-sector organizations that implement comparable automation levels (Vorontsova et al., 2025). The lower end of the Colombian range — 0.5 to 2 percentage points — reflects the expectation that automation depth in LAC HEIs will be limited to clerical and support functions in the near term, with managerial and specialized administrative roles largely unaffected.

5.3 Expected Heterogeneity

The calibration supports a strong prior that effects will be heterogeneous across institution type. Private institutions, which have higher baseline data infrastructure quality and greater administrative capacity, are expected to achieve efficiency gains at the upper end of the calibrated ranges. Public institutions — particularly small regional ones with limited IT capacity — are expected to cluster at the lower bound or below. The private-to-public ratio of expected effects, calibrated from Wang (2020) and Salas-Pilco et al. (2022), is in the range of 2 to 3.

This expected heterogeneity has a direct policy implication. If the ATT is driven primarily by private institutions, the aggregate efficiency gains from AI adoption in the Colombian system will be smaller than the literature suggests, and the distributional consequences will be regressive — with the largest gains accruing to institutions serving higher-income students. The empirical analysis is designed to test this prediction directly, with the group-time ATT structure of the Callaway-Sant'Anna estimator allowing separate identification of effects by institutional type and accreditation status.

5.4 Power Calculation and Survey Requirements

The calibration informs the minimum detectable effect (MDE) calculations for the survey design. Assuming a two-sided test at the 5 percent significance level with 80 percent power, detecting a dropout rate reduction of 1.5 percentage points — the lower bound of the calibrated range — requires approximately 90 treated institutions and 90 control institutions observed over at least six pre- and three post-adoption periods. The SNIES panel provides sufficient post-adoption observations for institutions that adopted AI tools before 2021. The primary survey is designed to recover adoption timing for a minimum of 90 institutions that adopted AI administrative tools between 2017 and 2022, ensuring adequate power at the lower bound of the expected effect distribution.

6. Conclusion

The adoption of artificial intelligence in university administration generates efficiency opportunities that are well-documented in the global literature and efficiency risks that are particularly acute in the Latin American institutional context. The gap between these two bodies of evidence — one descriptive and optimistic, the other also descriptive but cautionary — cannot be closed without causal identification. This paper proposes the identification strategy, data design, and calibration framework necessary to produce the first causal estimates of AI adoption effects in Colombian higher education.

The Colombian higher education system offers the conditions for credible identification: a comprehensive institutional panel in SNIES, variation in AI adoption timing generated in part by accreditation cycles and national digital transformation policy, and sufficient institutional diversity to detect heterogeneous effects across institution type. The primary survey instrument — designed to recover adoption timing at the institution level — is the binding data constraint, and the power calculations derived from the calibration exercise establish that it is feasible within a realistic budget.

The calibration exercise yields two empirically grounded conclusions that motivate the full analysis. First, the expected efficiency gains from AI adoption in Colombian HEIs — a 1.5 to 5 percentage point reduction in first-year dropout rates and an 8 to 18 percent reduction in administrative costs — are large enough to justify the policy investment in adoption support, conditional on the effects being confirmed causally. Second, the expected heterogeneity in these gains — with private institutions likely achieving effects two to three times larger than public institutions — indicates a distributional pattern that would make unrestricted adoption incentives regressive. Both predictions are testable with the proposed design.

The policy agenda implied by this analysis has three components: investing in the SNIES data infrastructure to support ongoing monitoring of AI adoption outcomes; developing AI governance frameworks for the Colombian higher education accreditation system that are calibrated to the institutional capacity of public universities; and designing adoption incentive programs — under CONPES 3975 and subsequent digital transformation policy — that explicitly condition support on governance requirements and prioritize equity-relevant functions. Policymakers at MEN and Minciencias should treat these as complementary rather than sequential investments. The governance frameworks are prerequisites for the incentive programs; the data infrastructure is a prerequisite for both.

It would be productive in future research to extend the analysis in two directions. First, examining the labor market effects of AI adoption on administrative staff — specifically, whether the staff-to-enrollment adjustments documented in the calibration reflect employment reductions or reallocation toward higher-skill administrative functions — would connect this research agenda directly to the broader literature on AI and labor market polarization. Second, a cross-country comparative analysis using similar identification strategies in Brazil, Mexico, and Chile would establish whether the Colombian findings generalize to the LAC region or reflect Colombia-specific institutional conditions.

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